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computer vision.

1) a) Main objectives of computer vision:

- It enables computers to capture, process and analyze images and videos to identify objects, patterns and scenes.
- Object detection and recognition.
- Image understanding and processing.
- Noise reducation.

b) Real world application of computere vission

- Medical image diagnosis
- sattlite image processing.

2) a) Feature extraction is the process of identifying and extracting important characteristics such as edges, corners colors from an image for analysis.

b) Object recognition identifies and classifies objects present in an image or video. it improve computer vision systems by enabling accurate decision making.

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3) a) Low level image processing:-
○) It deals with the basic image processing.
like noise reduction, image enhancement, filtering.

b) Mid level	High level.
○) image segmentation and feature extraction	Object recognition and scene understanding.
○) Product image features.	produces meaningful interpretation.

4) a) A pixel is the smallest unit of a digital image that store brightness or color information.

b) Image resolution:
○) Higher resolution contain more pixels, resulting better image quality and more detail.

5) a) Role of image sensor:

i) An image sensor converts light from a real world scene into electrical signals which are then converted into a digital image.

b) Two image acquisition devices.

i) Camera

ii) Webcam

6) a) Sampling converts a continuous image into discrete pixels.

b) Quantization ~~is~~ assigns intensity values to sample pixels creating a digital image.

7) b) Illumination provides the light required to capture image. Proper lighting improves image clarity, brightness and helps in accurate object detection and recognition.

A) ~~Pro~~ Perspective project is the process of projecting a 3D scene on to a 2D image plane, where objects that are farther away appear smaller than closer objects

8) a) Image Processing is the technique of performing operation on digital images to enhance their quality, remove noise,

b) Image Processing	Computer Vision
Enhances or modifies image	understaning and interprets image content
output is an improve image.	output is an understanding and meaningful data.

9) a) In healthcare computer vision is used to analyse X-rays and MRI scans to detect diseases such as tumors, fractures and other abnormalities.

b) In autonomous vehicles, computer vision is used for lane detection, traffic sign recognition, ~~pe~~ pedestrian detection and obstance avoidance enabling safe drive

10) a) Pixels are the smallest units of a digital image Each pixel store brightness or color information and together millions of pixels from a complete image. are the basic building blocks of digital images.